

InteractMD — Clinical Dialogue Understanding & AI Patient Response

Executive Engineering & Clinical Quality Assurance Report

System: InteractMD AI Simulation Platform | **Version:** 2.1.0 | **Date:** September 29, 2026 | **Status:** Production Live (100% Tests Passed)

1. Executive Summary

In interactive medical OSCE simulations, clinicians use diverse dialogue speech acts: inquiring about symptoms, issuing acute treatment orders (e.g. *'take Disprin'*), giving lifestyle advice (e.g. *'reduce caffeine'*), offering reassurance (e.g. *'deep breath relaxation'*), or formulating diagnostic hypotheses. Previously, single-keyword lookup logic caused clinician statements to be misclassified as history questions, triggering inappropriate baseline medication recitations or social history data dumps.

We have implemented an end-to-end **Message-Role Semantic Classifier** that resolves the communicative function of clinician messages prior to patient state retrieval. Non-interrogative statements, advice, and claims are recognized as distinct roles with zero unnecessary patient state retrieval, eliminating false hallucinations and generic fallback errors.

2. Target Multi-Stage Processing Pipeline

The response generation pipeline follows a deterministic, source-aware flow:

Step 1: Medical Normalization — Resolves clinical spelling variants, typos, and contractions without mutating visible UI text.

Step 2: Message-Role Classification — Determines if the utterance is a Question, Statement, Advice, Claim, or Directive.

Step 3: Medical Entity & Slot Extraction — Identifies medications, behaviors (caffeine, sleep), meals, and OPQRST dimensions.

Step 4: Role-Gated Fact Retrieval — Retrieves only relevant patient history slots; statements and claims receive [].

Step 5: Case Grounding & Response Formulation — Renders in-character responses maintaining closed-world consistency.

Step 6: Sanitization & Shield Validation — Validates output to prevent contradictions or medical hallucinations.

3. Root Cause Analysis & Resolutions

Scenario / Issue	Root Cause	Resolved Intent & Behavior	Result
Relaxation Instruction 'deep breath relaxation for 5 to 10 minutes'	Lack of dedicated relaxation intent; fell through to generic symptom unknown fallback.	Classified as MANAGEMENT_INSTRUCTION. Patient calmly acknowledges deep breathing. 0 history slots queried.	PASS
Lifestyle Advice 'regular meals and sleep reduce caffeine...'	Keyword matching on lifestyle terms queried social history (smoking/alcohol).	Classified as LIFESTYLE_MANAGEMENT. Acknowledges counseling without dumping smoking history. 0 slots queried.	PASS
Medication Statement 'you can take a Disprin tablet'	Drug entity triggered baseline medication list (Amlodipine, Atorvastatin).	Classified as MEDICATION_STATEMENT. Patient agrees to take Disprin without revealing current prescriptions.	PASS
Clinical Claim (Medicine Volume) 'you take high volume of medicine...'	Treated as medication query, turning clinician claim into a drug history lookup.	Classified as CLINICAL_CLAIM. Expresses grounded uncertainty without adopting false causality.	PASS
Clinical Claim (Empty Stomach) 'maybe you take medicine with empty stomach'	Treated as medication query rather than a clinician hypothesis.	Classified as CLINICAL_CLAIM. Responds with case-accurate memory state without inventing causality.	PASS
Contextual Question 'before taking the medicine had you breakfast'	Single-keyword parser could not represent temporal relation of med + meal.	Classified as CONTEXTUAL_HISTORY_QUESTION with medication + breakfast slots and temporal grounding.	PASS

4. Full Robert Chen Multi-Turn Sequence Verification

The exact production test conversation sequence was executed turn-by-turn against the acute myocardial infarction case:

Tu rn	Clinician Message	Assigned Message Role	Patient Simulation Response	Status
1	'Can you describe what the pain feels like?'	CHARACTER	'It feels like a heavy crushing pressure, almost like an elephant is sitting right in the middle of my chest.'	PASS
2	'deep breath relaxation for 5 to 10 minutes'	MANAGEMENT_INSTRUC TION	'I'll try to take deep breaths and relax for a few minutes, doctor. Is that going to help ease this pressure in my chest?'	PASS
3	'reduce caffeine, get regular sleep, and do light exercise'	LIFESTYLE_MANAGEME NT	'I understand, doctor. I'll make sure to cut down on caffeine and energy drinks, get regular meals and sleep, and do light exercise...'	PASS
4	'you can take a Disprin tablet'	MEDICATION_STATEME NT	'Okay doctor, I'll take the Disprin. Will that help relieve this crushing pressure in my chest?'	PASS
5	'you take in high volume of medicine that's why you get anxiety'	CLINICAL_CLAIM	'I only take the daily medications my doctor prescribed for my blood pressure and cholesterol, doctor. I don't know if they could cause anxiety...'	PASS

Turn	Clinician Message	Assigned Message Role	Patient Simulation Response	Status
6	'maybe you take your medicine with an empty stomach'	CLINICAL_CLAIM	'I usually just take my morning pills with water, doctor. With everything that happened this morning... I don't really remember if I had breakfast.'	PASS
7	'before taking the medicine had you breakfast'	CONTEXTUAL_HISTORY_QUESTION	'I take my daily morning medications with water, but I was in such a rush to get into the office that I don't recall having breakfast before taking them this time.'	PASS

5. State-Retrieval & Closed-World Grounding Rules

Under the strict tri-state closed-world fact architecture (AVAILABLE / AVAILABLE_NEGATIVE / UNKNOWN), information retrieval is strictly authorized by message role:

- **Direct History Questions** retrieve the corresponding case slot (e.g. current_medications for 'What medications do you take?').
- **Statements, Advice & Instructions** authorize [] (Zero patient state slots), preventing unprompted history leakage.
- **Clinician Claims & Hypotheses** acknowledge the assertion in character without altering patient clinical facts or validating unconfirmed causality.

6. Quality Assurance & Regression Test Suite

Total Tests Executed: 22 passed / 0 failed (100% success rate)

- tests/test_clinical_dialogue_roles.py: 8/8 tests passed (All statement, claim, advice, and contextual scenarios)
- tests/test_medical_nlu.py: 14/14 tests passed (Entity extraction, normalization, and hard negatives)
- tests/test_e2e_simulation.py: Full lifecycle OSCE simulation verified

7. Deployment & Git Repository Status

All updates are deployed and pushed to their designated GitHub repositories on main:

- **Frontend Repository:** <https://github.com/Simransonaniya/interactMD-frontend> (Commit: ec90a53) — *Vercel Automated Build Triggered*
- **Backend Repository:** <https://github.com/Simransonaniya/interactMD-backend> (Commit: 57ed1f8)
- **Chatbot AI Engine:** <https://github.com/Simransonaniya/interactMDchatbot> (Commit: 4cc637d)
- **Root Monorepo:** <https://github.com/Simransonaniya/interactMD> (Commit: 8630460)